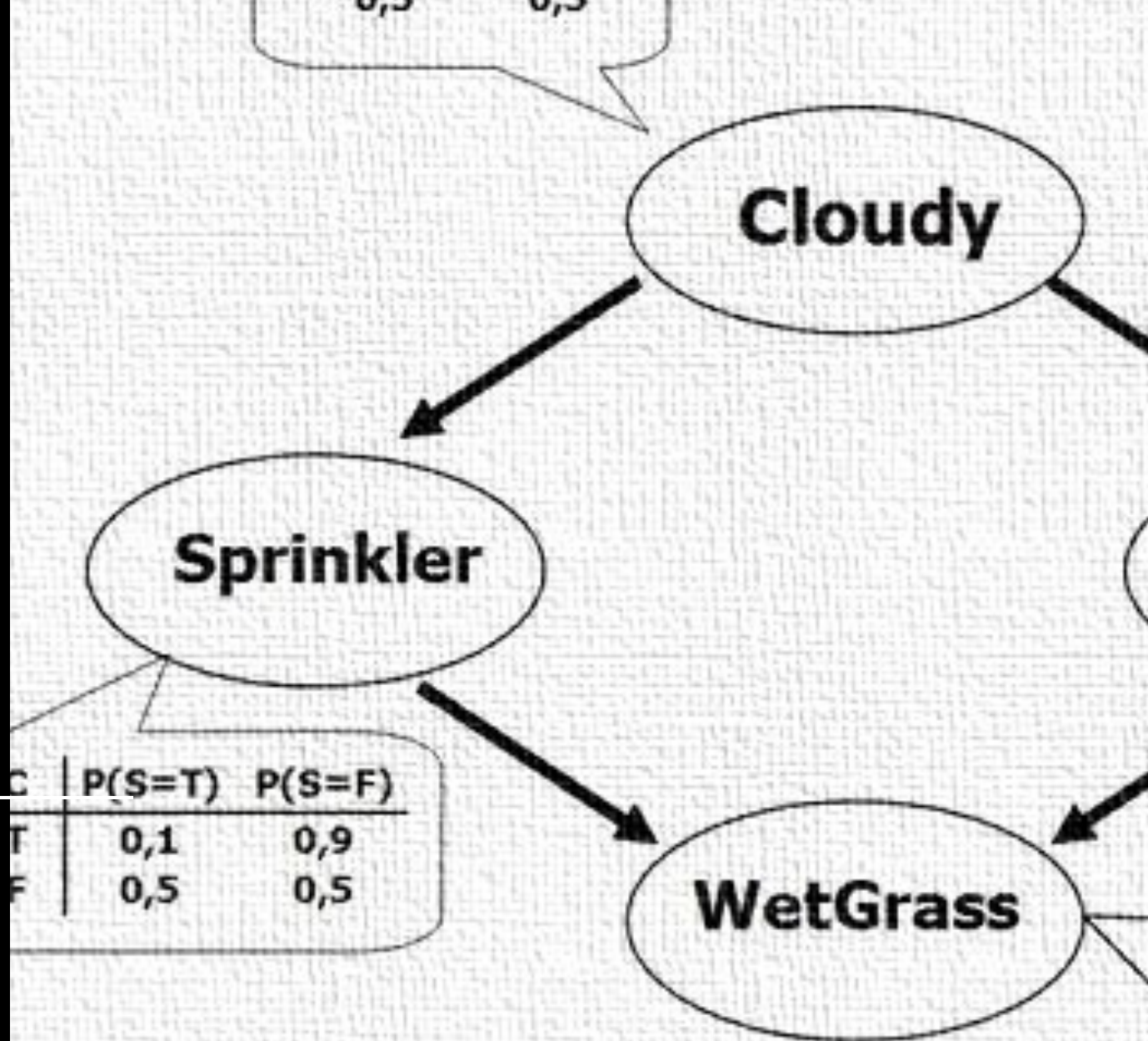


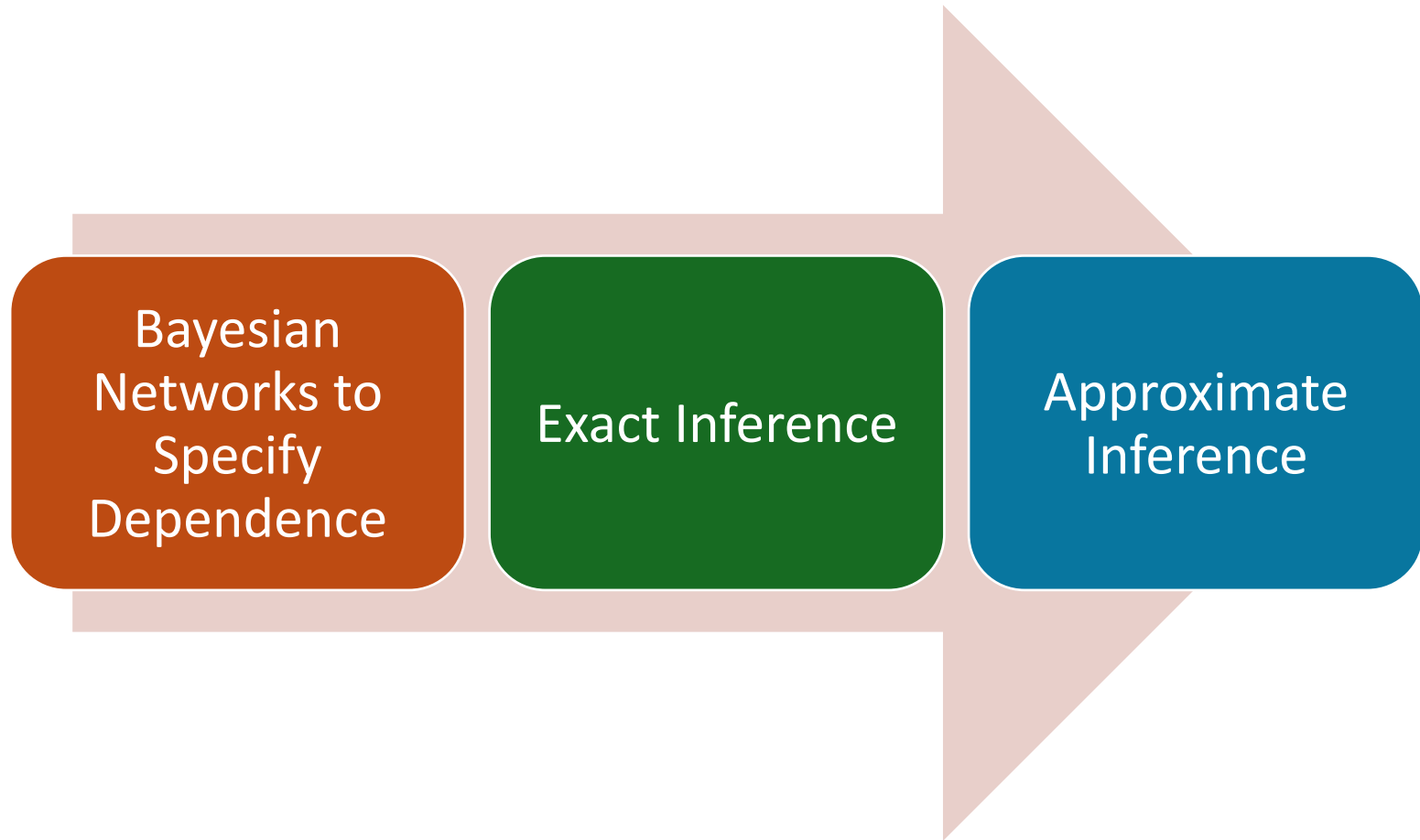
Probabilistic
Reasoning
(Bayesian Networks
)



Probability Theory Recap

- Notation: Prob. of an event $P(X = x) = P(x)$
Prob. distribution $\mathbf{P}(X) = \langle P(X = x_1), P(X = x_2), \dots, P(X = x_n) \rangle$
- Product rule $P(x, y) = P(x|y)P(y)$
- Chain rule
$$\begin{aligned}\mathbf{P}(X_1, X_2, \dots, X_n) &= \mathbf{P}(X_1)\mathbf{P}(X_2|X_1)\mathbf{P}(X_3|X_1, X_2) \dots \\ &= \prod_{i=1}^n \mathbf{P}(X_i|X_1, \dots, X_{i-1})\end{aligned}$$
- Conditional probability $P(x|y) = \frac{P(x, y)}{P(y)} = \alpha P(x, y)$
- Marginal distribution given $\mathbf{P}(X, Y)$
 $\mathbf{P}(X) = \sum_y \mathbf{P}(X, y)$ (called marginalizing out Y)
- Independence
 - $X \perp\!\!\!\perp Y$: X, Y are independent (written as $X \perp\!\!\!\perp Y$) if and only if:
$$\forall x, y: P(x, y) = P(x)P(y)$$
 - $X \perp\!\!\!\perp Y|Z$: X and Y are conditionally independent given Z if and only if:
$$\forall x, y, z: P(x, y|z) = P(x|z)P(y|z)$$

Contents



Cloudy

Sprinkler

Rain

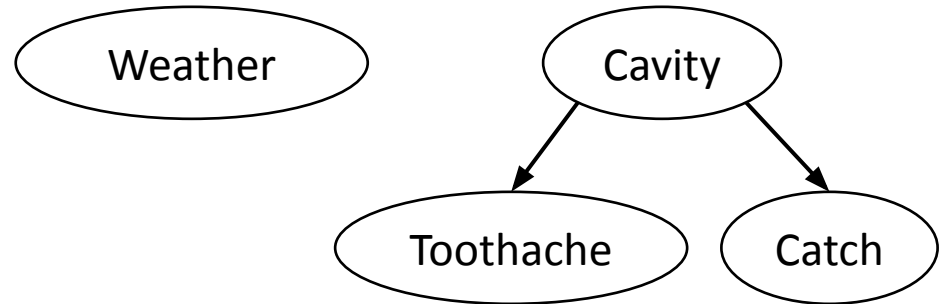
Bayesian Networks

Modeling a Joint Distribution

C	P(S=T)	P(S=F)
T	0,1	0,9
F	0,5	0,5

S	R	P(W=)
T	T	0,9
T	F	0,9
F	T	0,9
F	F	0,0

Bayesian Networks (aka Belief Networks)



A type of graphical model.



A way to specify dependence between random variables.



A compact specification of a full joint probability distribution.

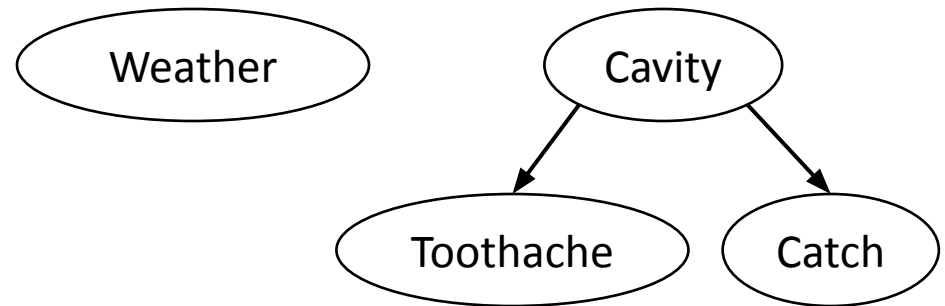


A general and important model to reason with uncertainty in AI.

Structure of Bayesian Networks

Nodes: Random variables

- Can be assigned (observed) or unassigned (unobserved)



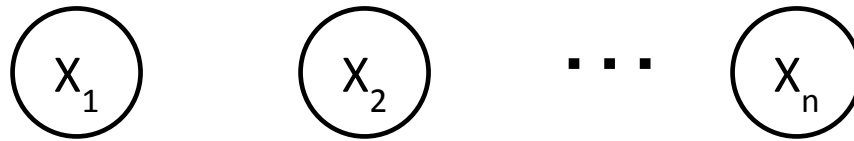
Arcs: Dependencies

- An arrow from one variable to another indicates direct influence.
- Show independence
 - *Weather* is independent of the other variables (no connection).
 - *Toothache* and *Catch* are conditionally independent given *Cavity* (directed arc).
- Must form a directed *acyclic* graph (DAG)

A network with all random variables assigned represents a **state of the system**.

Example: N independent coin flips

Complete independence: no interactions between coin flips



$$P(X_1, X_2, \dots, X_n) = P(X_1)P(X_2) \dots P(X_n) = \prod_{i=1}^n P(X_i)$$

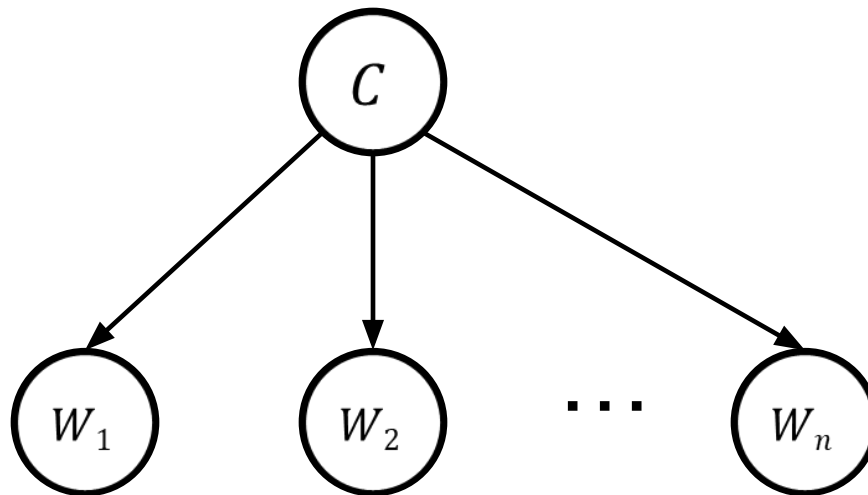
Joint probability
distribution

Example: Naïve Bayes spam filter

Random variables:

- C : message class (spam or not spam)
- $W_1, \dots, W_n$: presence or absence of words comprising the message

Words depend on the class, but they are modeled conditional independent of each other given the class (= no direct connection between words).

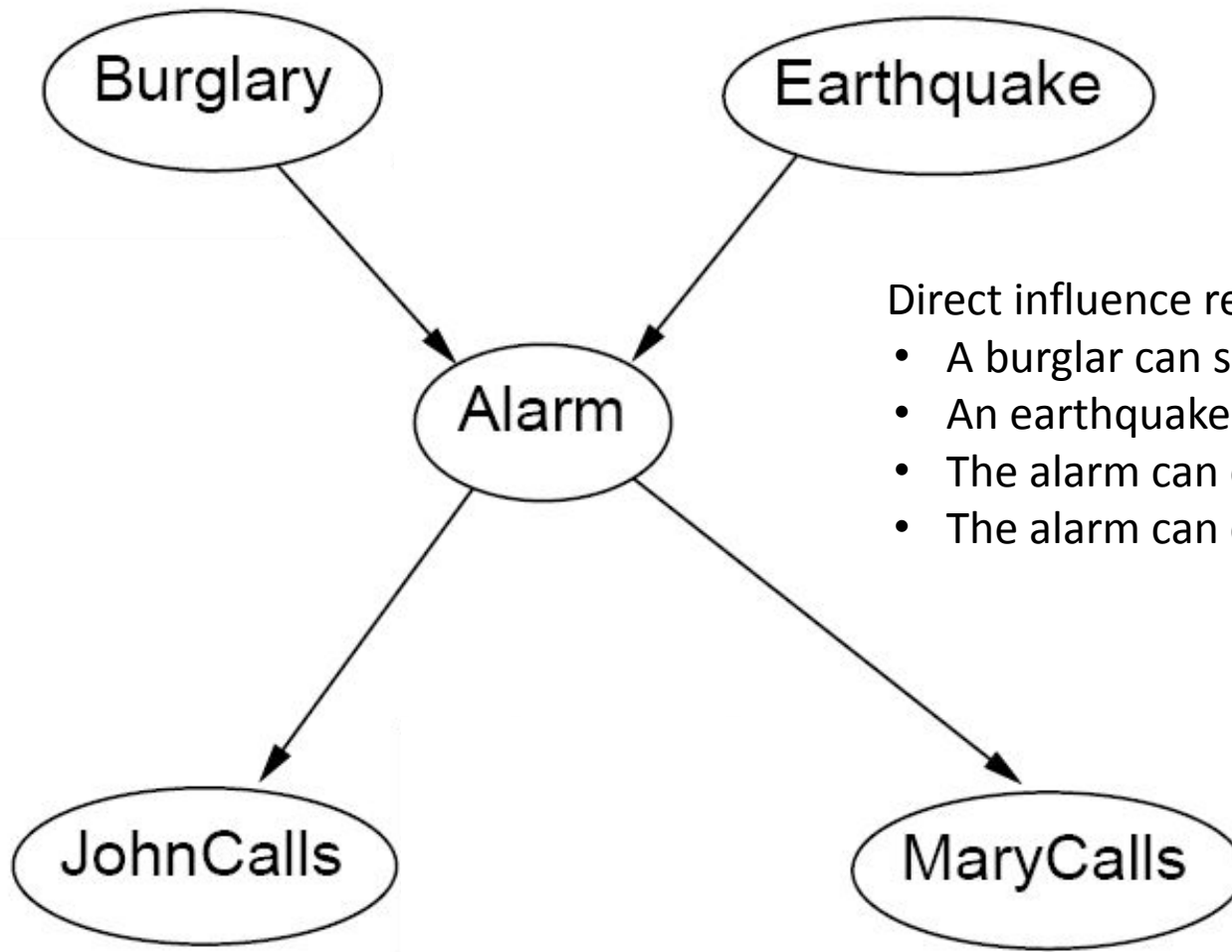


$$P(W_1, W_2, \dots, W_n | C) = P(W_1 | C) P(W_2 | C) \dots P(W_n | C)$$

Example: Burglar Alarm

- **Description:** I have a burglar alarm that is sometimes set off by minor earthquakes. My two neighbors, John and Mary, promised to call me at work if they hear the alarm
- Example inference task: suppose Mary calls and John doesn't call. What is the probability of a burglary?
- What are the random variables?
 - Burglary, Earthquake, Alarm, JohnCalls, MaryCalls
- What are the direct influence relationships?
 - A burglar can set off the alarm
 - An earthquake can set off the alarm
 - The alarm can cause Mary to call
 - The alarm can cause John to call

Example: Burglar Alarm as a Network



Direct influence relationships:

- A burglar can set off the alarm
- An earthquake can set off the alarm
- The alarm can cause Mary to call
- The alarm can cause John to call

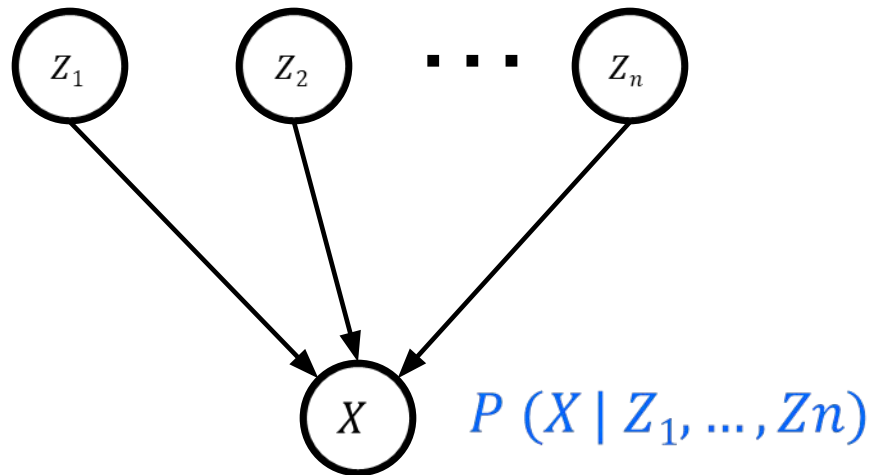
Parameters: Conditional probability tables

The full joint distribution, can be broken down into *conditional* distribution for each node given its parents:

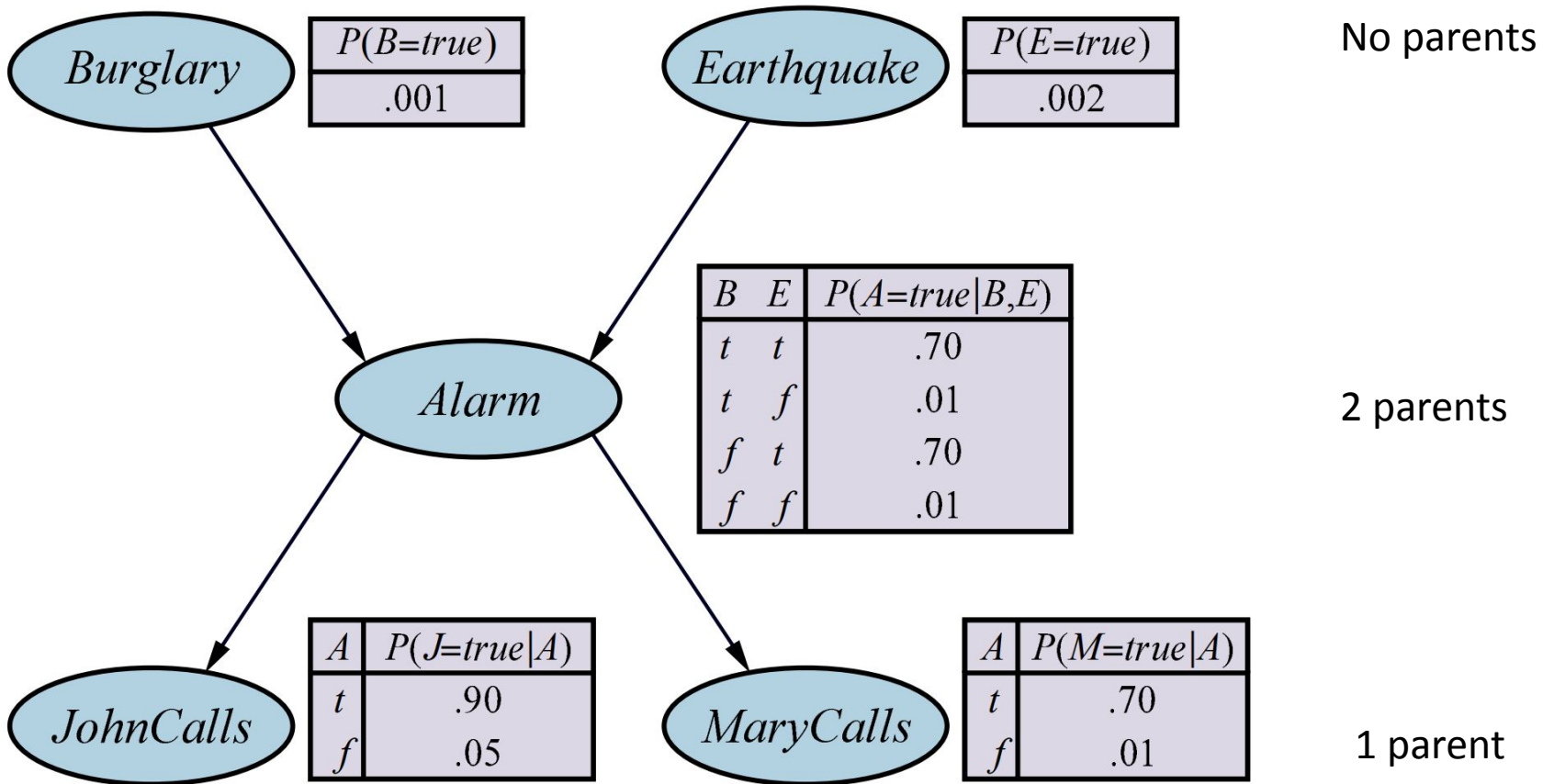
$$P(X | \text{Parents}(X))$$

These distributions are stored in conditional probability tables (CPTs)

Example:



Example: Burglar Alarm with CPTs

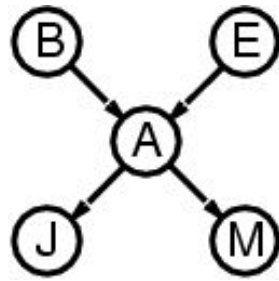


The Joint Probability Distribution

- For each node X_i , we know $P(X_i | Parents(X_i))$
- How do we get the full joint distribution $P(X_1, \dots, X_n)$?
- Using chain rule, but only depends on parents:

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | Parents(X_i))$$

- Example:

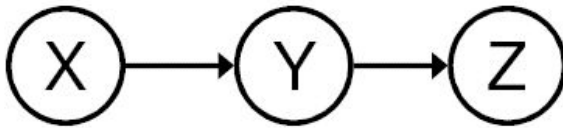


Construct
following
arrows

$$P(J, M, A, B, E) = P(B) P(E) P(A | B, E) P(J | A) P(M | A)$$

Dependence

- Example: *causal chain*



X: Low pressure

Y: Rain

Z: Traffic

- Are X and Z independent?

1. Conditioning: $P(X, Y, Z) = P(X)P(Y|X)P(Z|Y)$

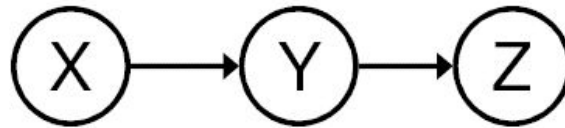
2. Marginalize over y: $P(X, Z) = \sum_y P(X)P(y|X)P(Z|y)$
 $= P(X) \sum_y P(Z|y)P(y|X) \neq P(X)P(Z)$

We are not
interested in y.

**X and Z are not
independent!**

Conditional Independence

- Example: *causal chain*



X: Low pressure

Y: Rain

Z: Traffic

- Is Z independent of X given Y?

1. Conditioning:
$$P(X, Z|Y) = \frac{P(X, Y, Z)}{P(Y)} = \frac{P(X)P(Y|X)P(Z|Y)}{P(Y)}$$

2. Bayes' rule:
$$= \frac{P(X) \frac{P(X|Y)P(Y)}{P(X)} P(Z|Y)}{P(Y)} = P(X|Y)P(Z|Y)$$

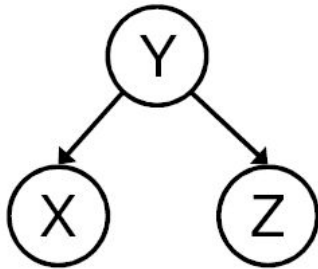
= Definition of
conditional
independence



X and Z are conditionally
independent given Y

Conditional Independence cont.

- *Common cause*



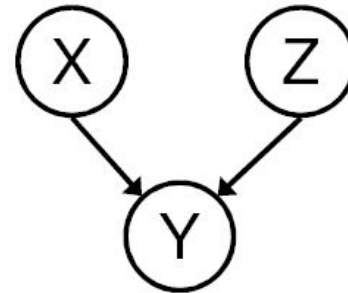
Y: Project due

X: Newsgroup
busy

Z: Lab full

- Are X and Z independent?
 - No
- Are they conditionally independent given Y?
 - Yes

- *Common effect*



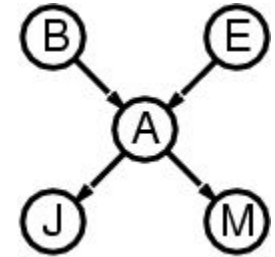
X: Raining

Z: Ballgame

Y: Traffic

- Are X and Z independent?
 - Yes
- Are they conditionally independent given Y?
 - No

Compactness



- For a network with n Boolean variables, the full joint distribution requires $O(2^n)$ probabilities.

Example: Burglary network

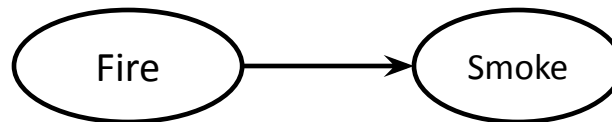
Complete joint probability specification $2^5 - 1 = 31$

- If each variable X_i has at most k Boolean parents, then each conditional probability table (CPT) has at most 2^k rows.
- The CPTs for all n nodes contain then at most $O(n \times 2^k)$ probabilities.
- This reduces the complexity from exponential to linear in n and makes it very compact!
- Example: Burglary network
Using CPTs: $1 + 1 + 4 + 2 + 2 = 10$ probabilities

Constructing Bayesian Networks

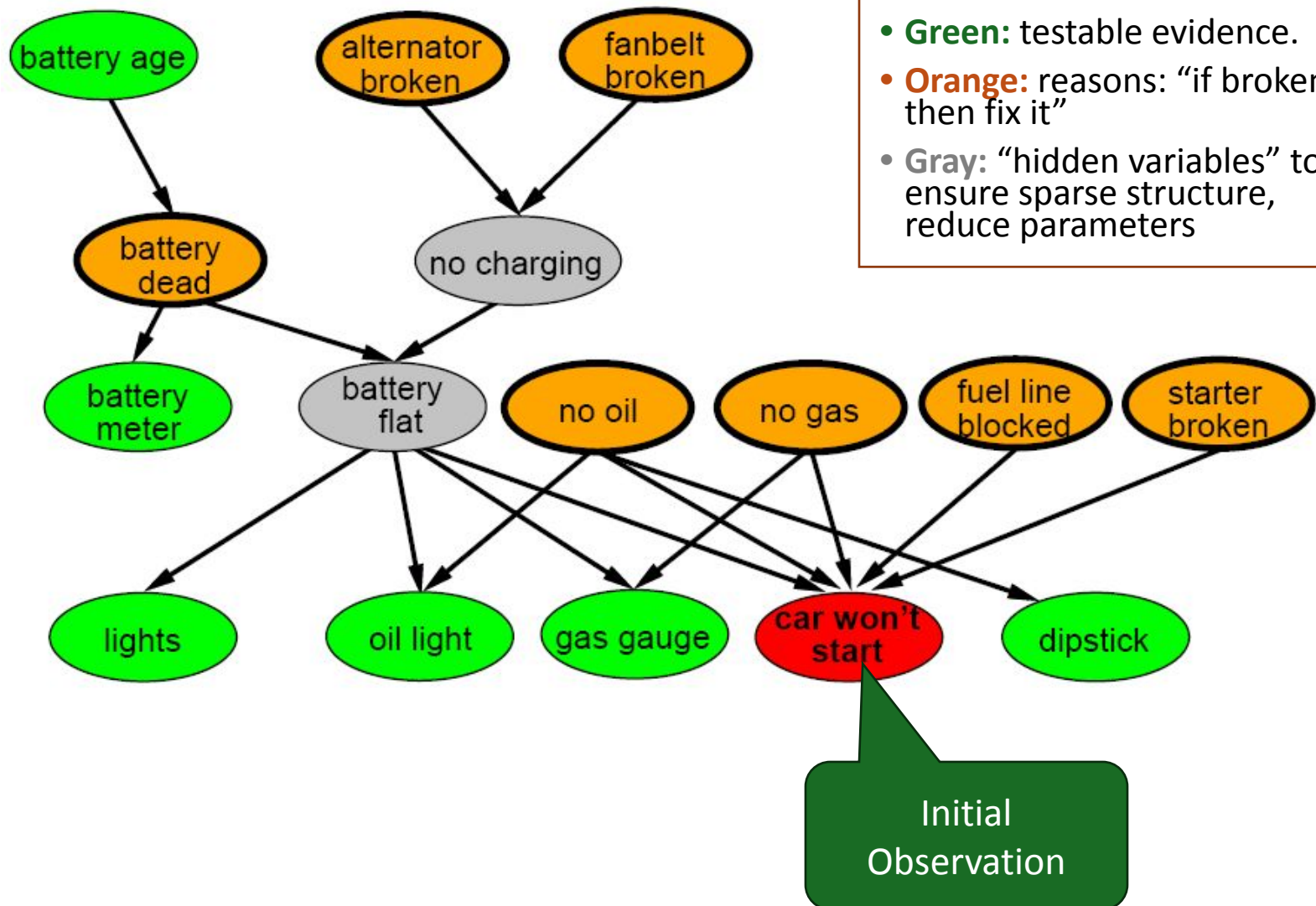
1. Choose an ordering of variables $X_1, \dots, X_n$
2. For $i = 1$ to n
 - add X_i to the network
 - select parents from $X_1, \dots, X_{i-1}$ such that
$$P(X_i \mid \text{Parents}(X_i)) = P(X_i \mid X_1, \dots, X_{i-1})$$
that is, add a connection only from nodes it directly depends on.

Note: There are many ways to order the variables. Networks are typically constructed by domain experts with causality in mind. E.g., Fire causes Smoke:

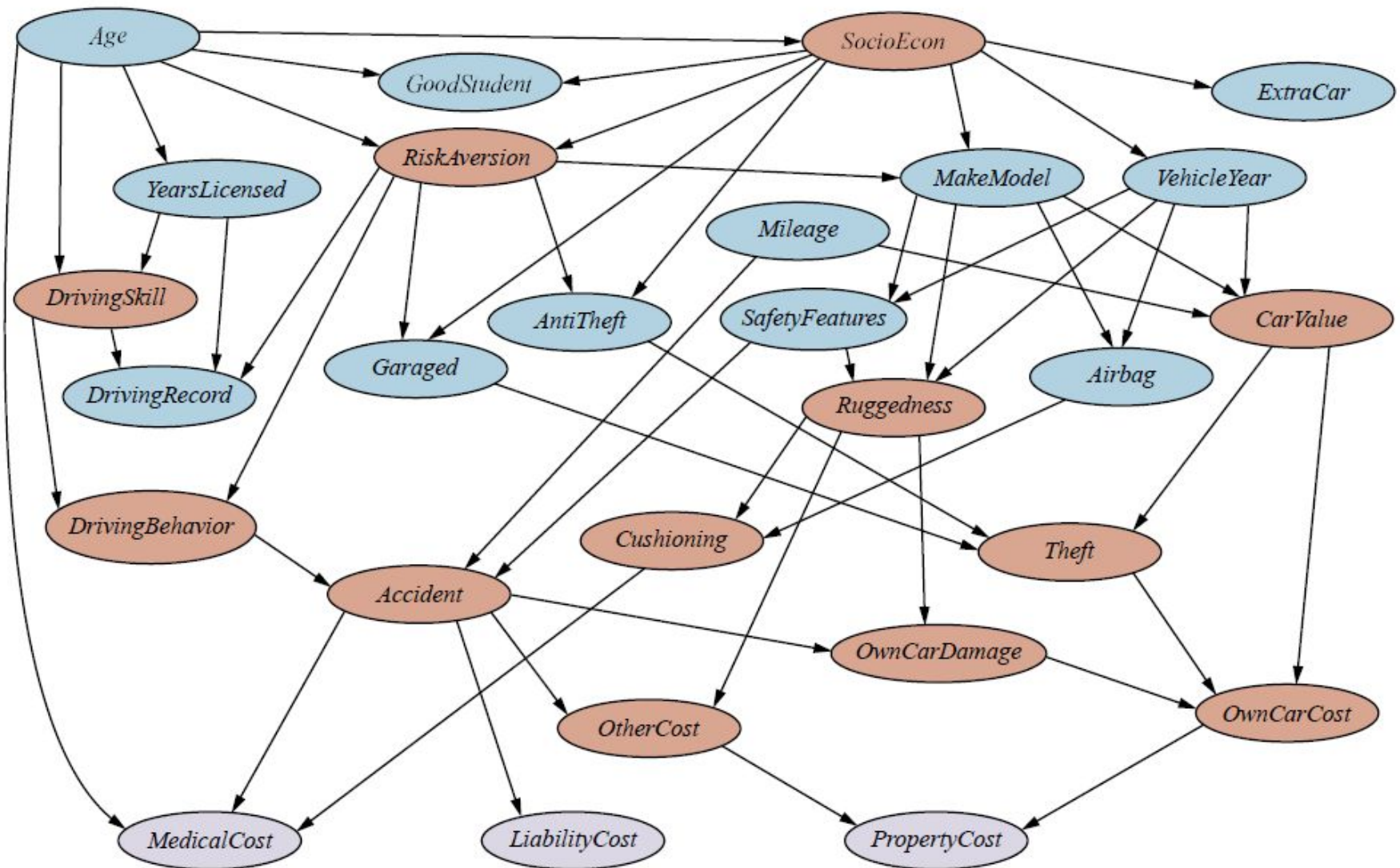


The network resulting from causal ordering is typically sparse and conditional probabilities are easier to judge because they represent causal relationships.

A more realistic Bayes Network: Car diagnosis

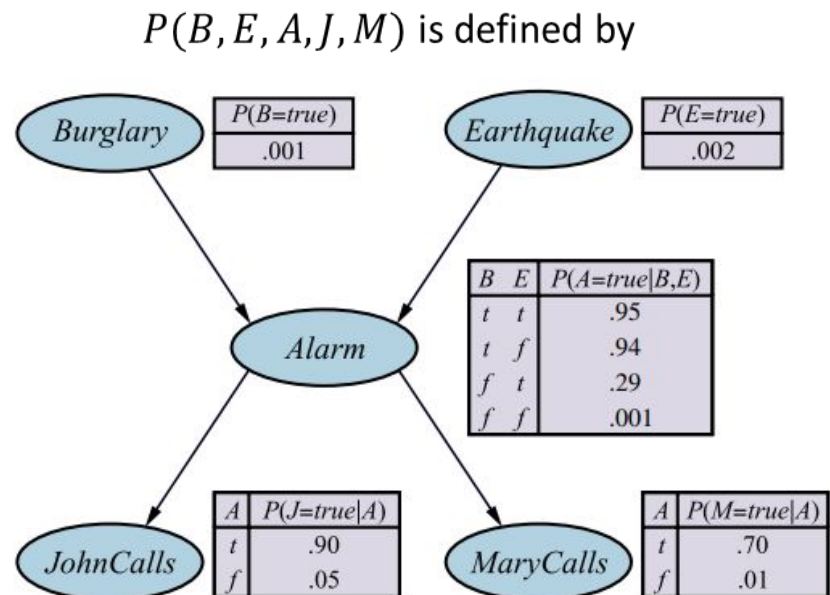


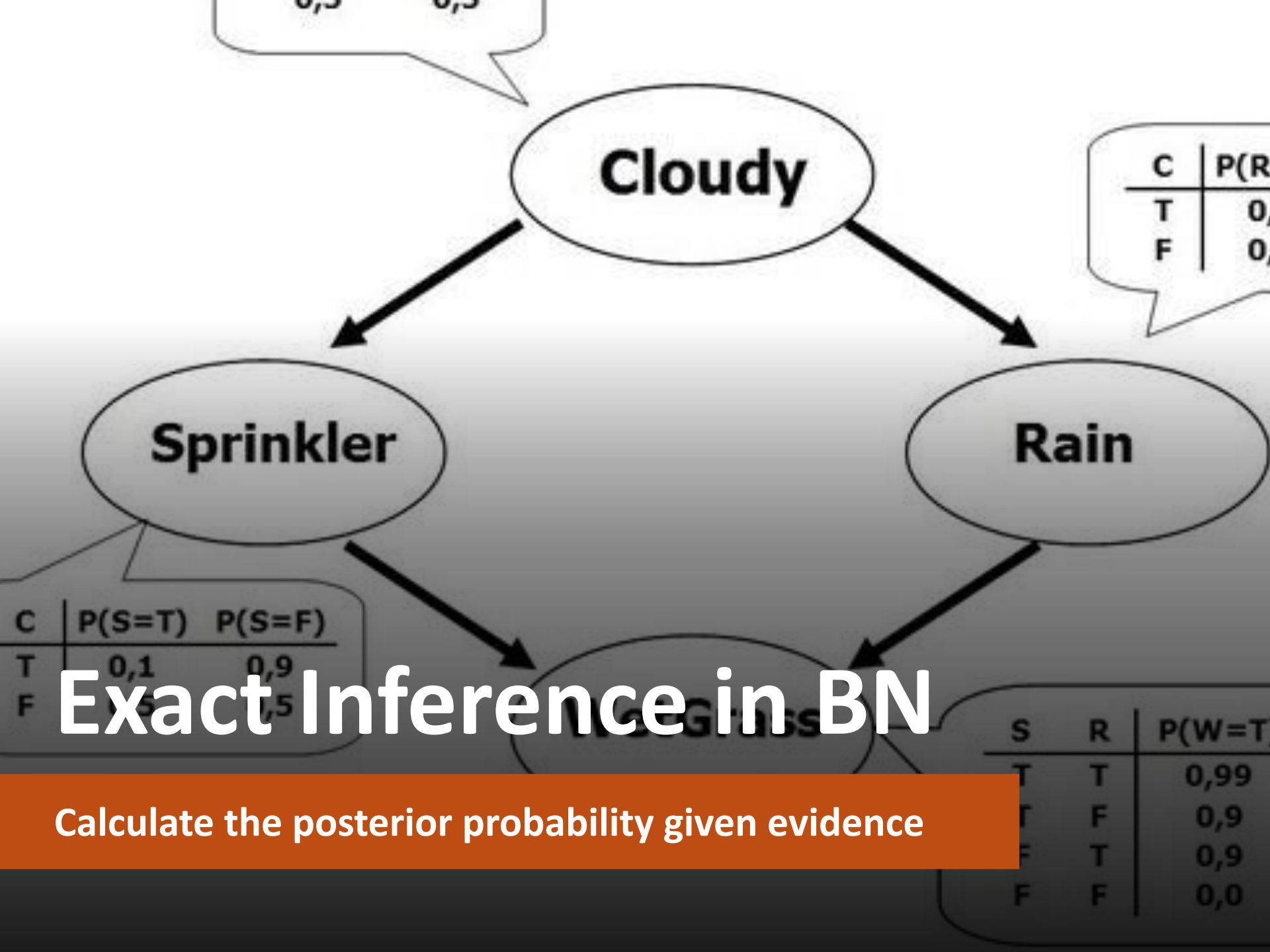
Car insurance: Cost is affected by many factors



Summary

- Bayesian networks provide a **natural representation for joint probabilities** used to calculate conditional probabilities needed for inference (prediction).
- **Independence** and **conditional independence** (induced by causality) reduces the number of needed parameters and creates a compact network.
- Representation
 - Topology (nodes and edges)
 - Conditional probability tables
 - Typically easy for domain experts to construct





Exact Inference in BN

Calculate the posterior probability given evidence

Exact Inference

Goal

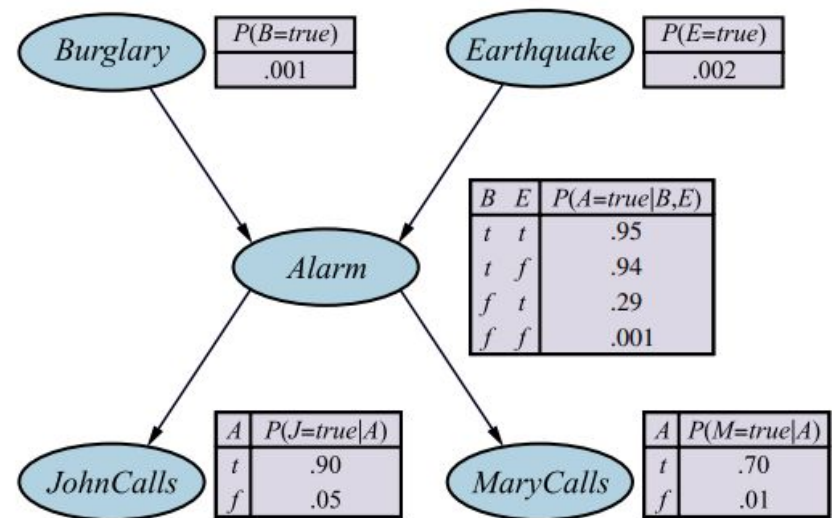
- Query variables: X
- Evidence (observed) variables: $E = e$
- Set of unobserved variables: Y
- Calculate the probability of X given e .

If we know the full joint distribution $P(X, E, Y)$, we can infer X by:

$$P(X|E = e) = \frac{P(X, e)}{P(e)} = \frac{\sum_y P(X, e, y)}{P(e)} \propto \sum_y P(X, e, y)$$

Sum over values of unobservable variables = marginalizing them out.

Exact inference: Example – Calculation



Assume we can observe being called and the two variables have the values j and m .
We want to know the probability of a burglary.

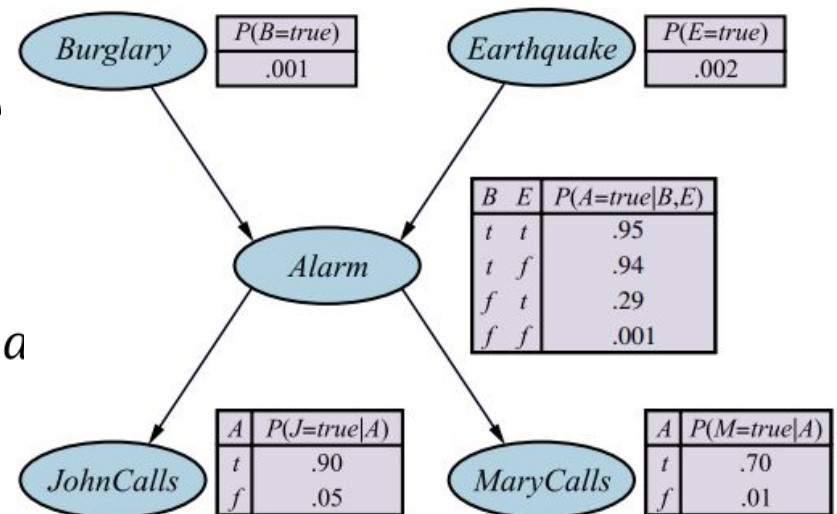
Query: $P(B | j, m)$ with unobservable variables: Earthquake E , Alarm A

$$\begin{aligned}
 P(b|j, m) &= \frac{P(b, j, m)}{P(j, m)} \propto \sum_e \sum_a P(b, e, a, j, m) \\
 &= \sum_e \sum_a P(b)P(e)P(a|b, e)P(j|a)P(m|a) \\
 &= P(b) \sum_e P(e) \sum_a P(a|b, e) P(j|a)P(m|a)
 \end{aligned}$$

Full joint
probability and
marginalize over E
and A

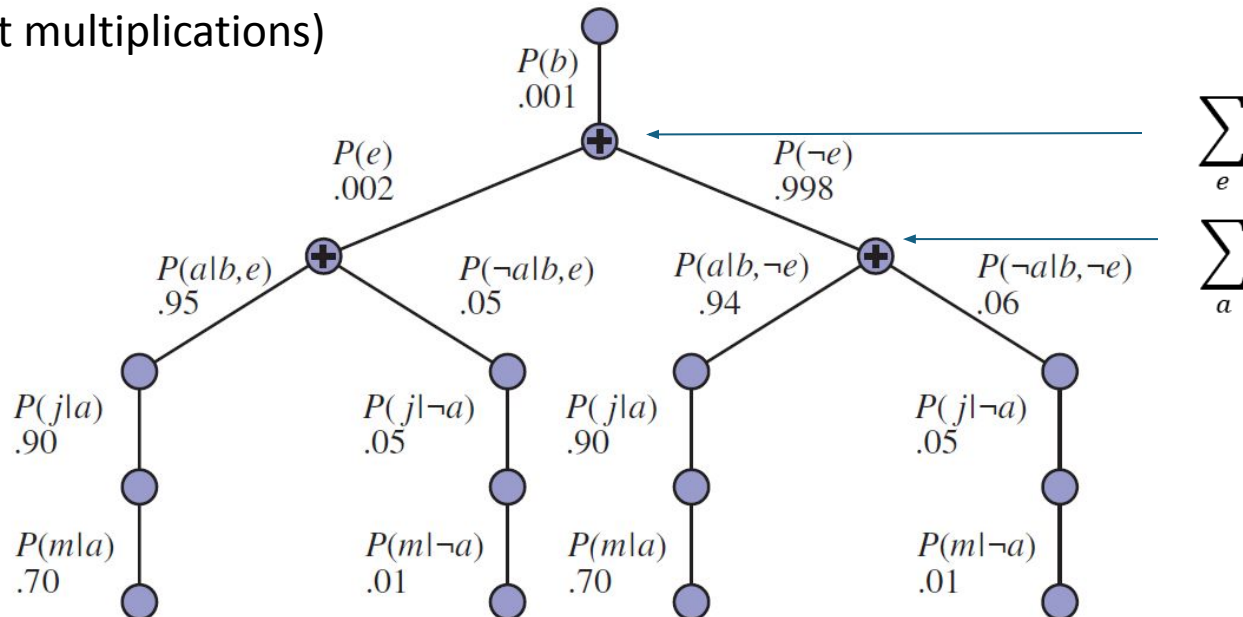
Exact inference: Example – Evaluation Tree

$$P(b|j, m) \propto P(b) \sum_e P(e) \sum_a P(a|b, e) P(j|a) P(m|a)$$

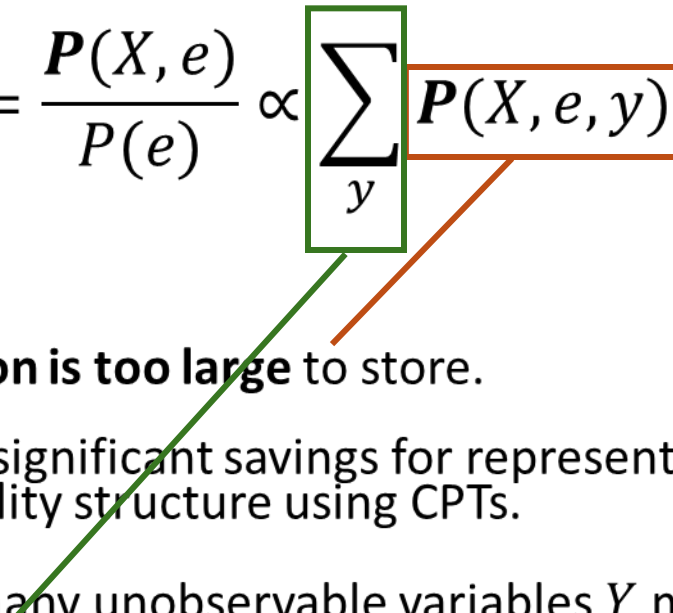


Implementation using an evaluation tree and CPTs

(lines represent multiplications)



Issues with Exact Inference in AI

$$P(X|E = e) = \frac{P(X, e)}{P(e)} \propto \sum_y P(X, e, y)$$


Problems

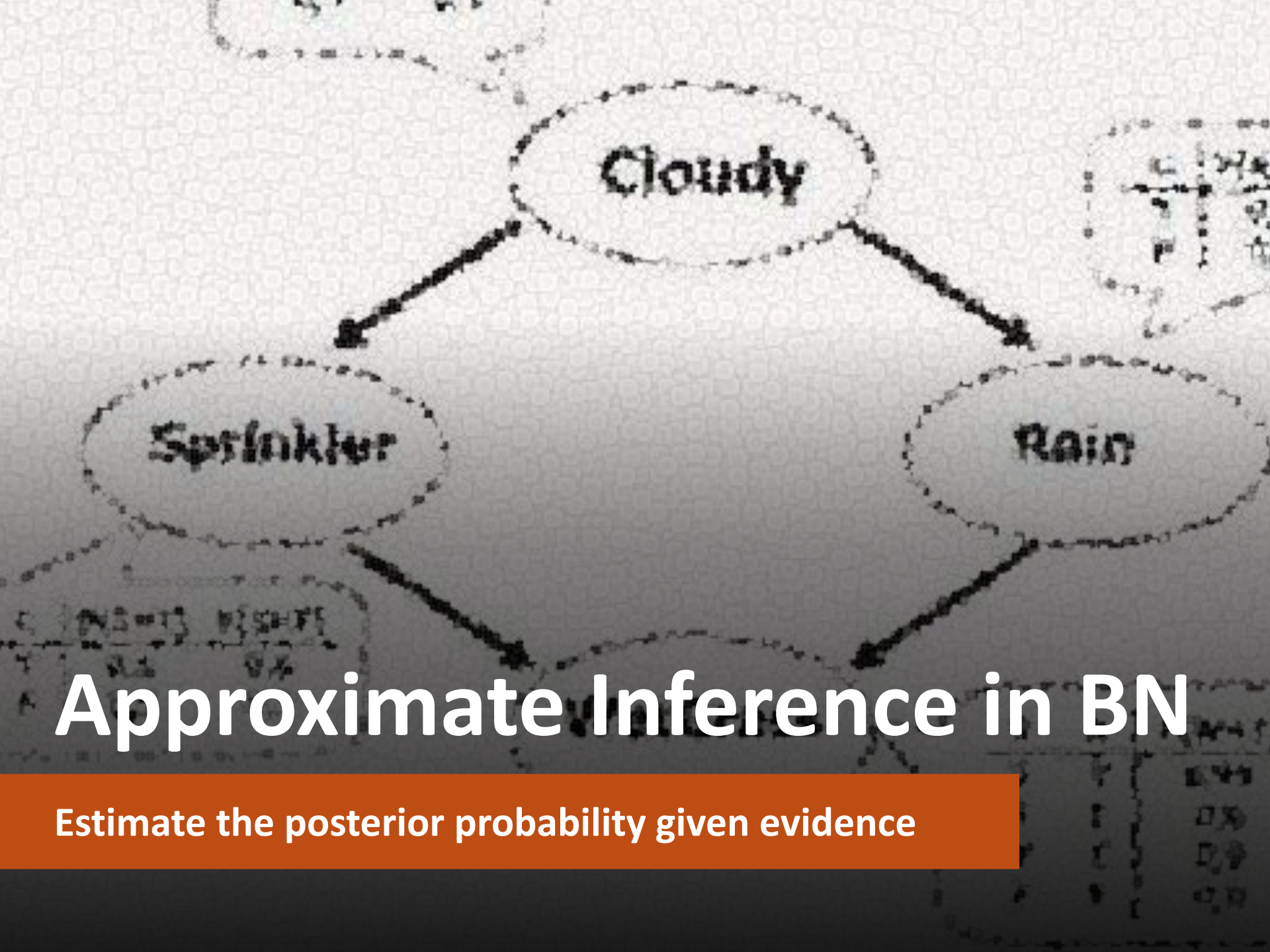
1. **Full joint distribution is too large** to store.

Bayes nets provide significant savings for representing the conditional probability structure using CPTs.

2. Marginalizing out many unobservable variables Y may involve **too many summation terms**.

This summation is called **exact inference by enumeration**. Unfortunately, it does not scale well (#p-hard).

In praxis, **approximate inference by sampling** is used.



Approximate Inference in BN

Estimate the posterior probability given evidence

Bayesian Networks as a Generative Models



Bayesian networks can be used as ***generative models***.



Generative models allow us to efficiently **generate samples** from the joint distribution.

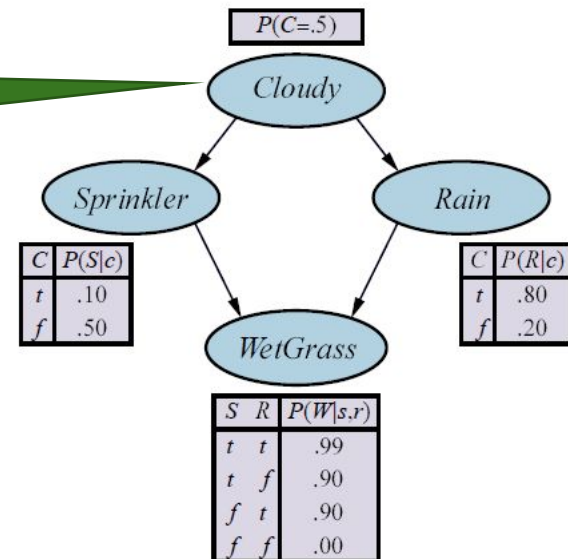


Idea: Generate samples from the network to estimate joint and conditional probability distributions using Monte Carlo methods.

Prior-Sample Algorithm

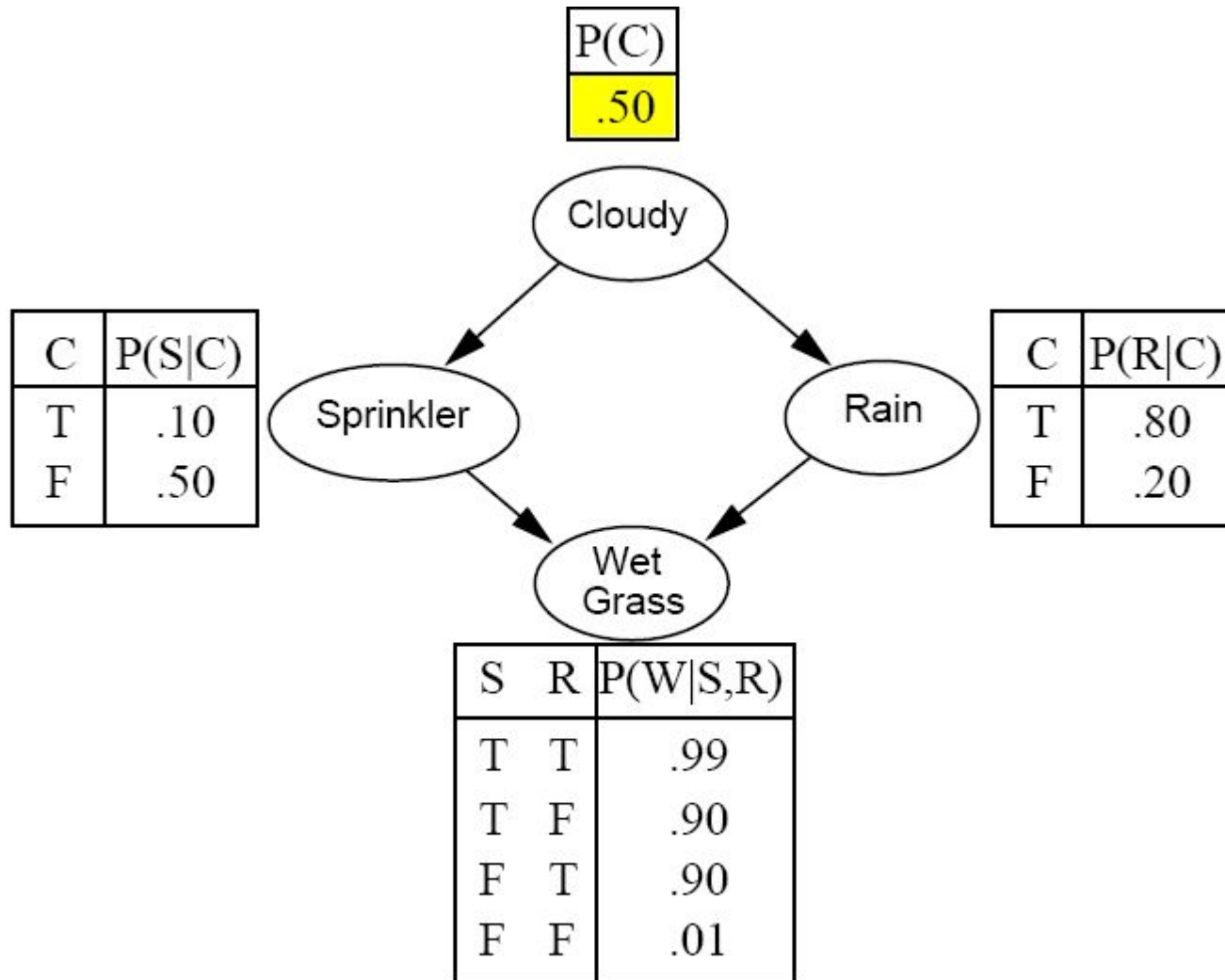
```
function PRIOR-SAMPLE( $bn$ ) returns an event sampled from the prior specified by  $bn$   
inputs:  $bn$ , a Bayesian network specifying joint distribution  $\mathbf{P}(X_1, \dots, X_n)$   
  
 $\mathbf{x} \leftarrow$  an event with  $n$  elements  
for each variable  $X_i$  in  $X_1, \dots, X_n$  do  
     $\mathbf{x}[i] \leftarrow$  a random sample from  $\mathbf{P}(X_i \mid \text{parents}(X_i))$   
return  $\mathbf{x}$ 
```

Order is important! We need to start with the random variables that have no parents.



Variable Order

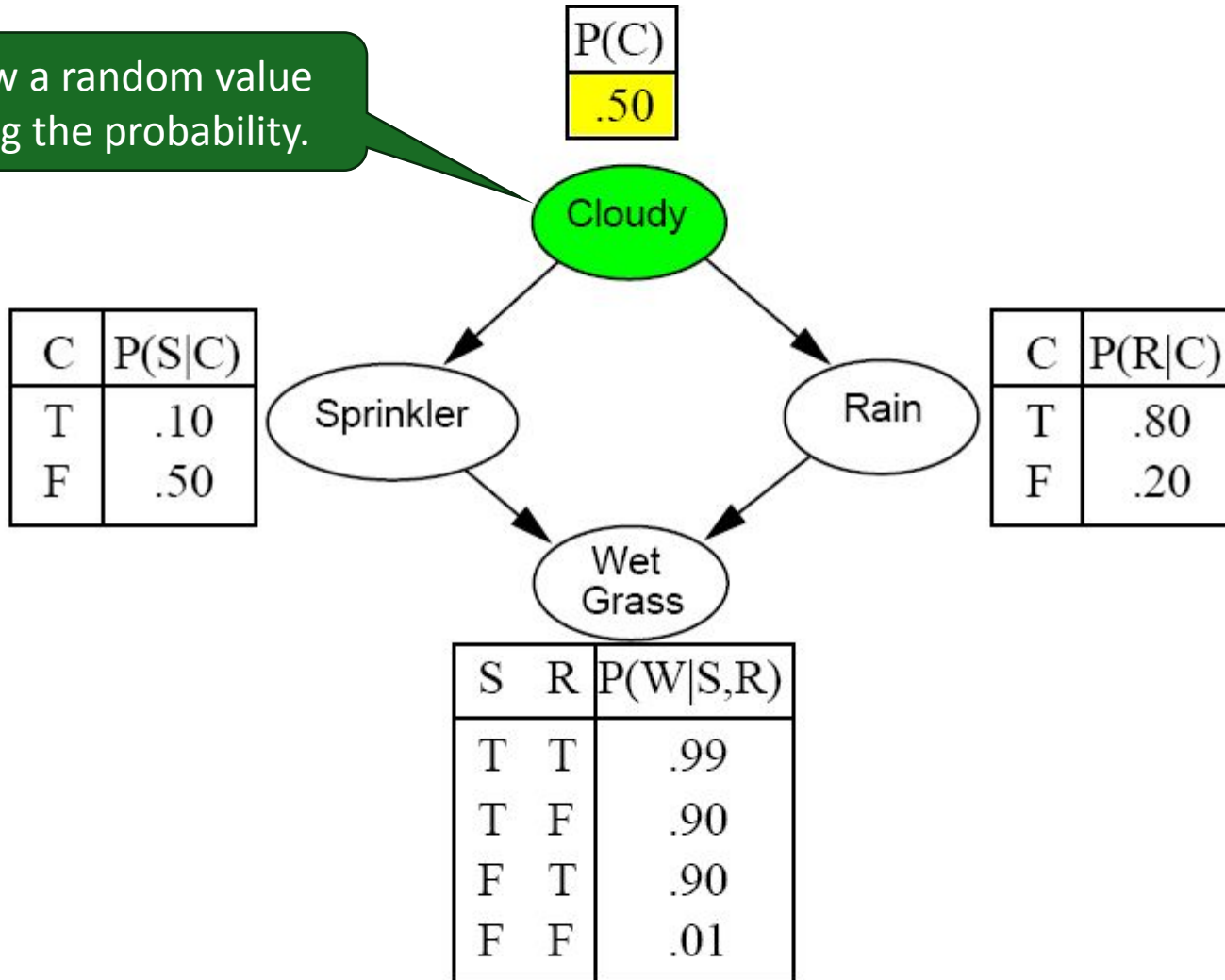
Example: Sampling from a Bayesian Network



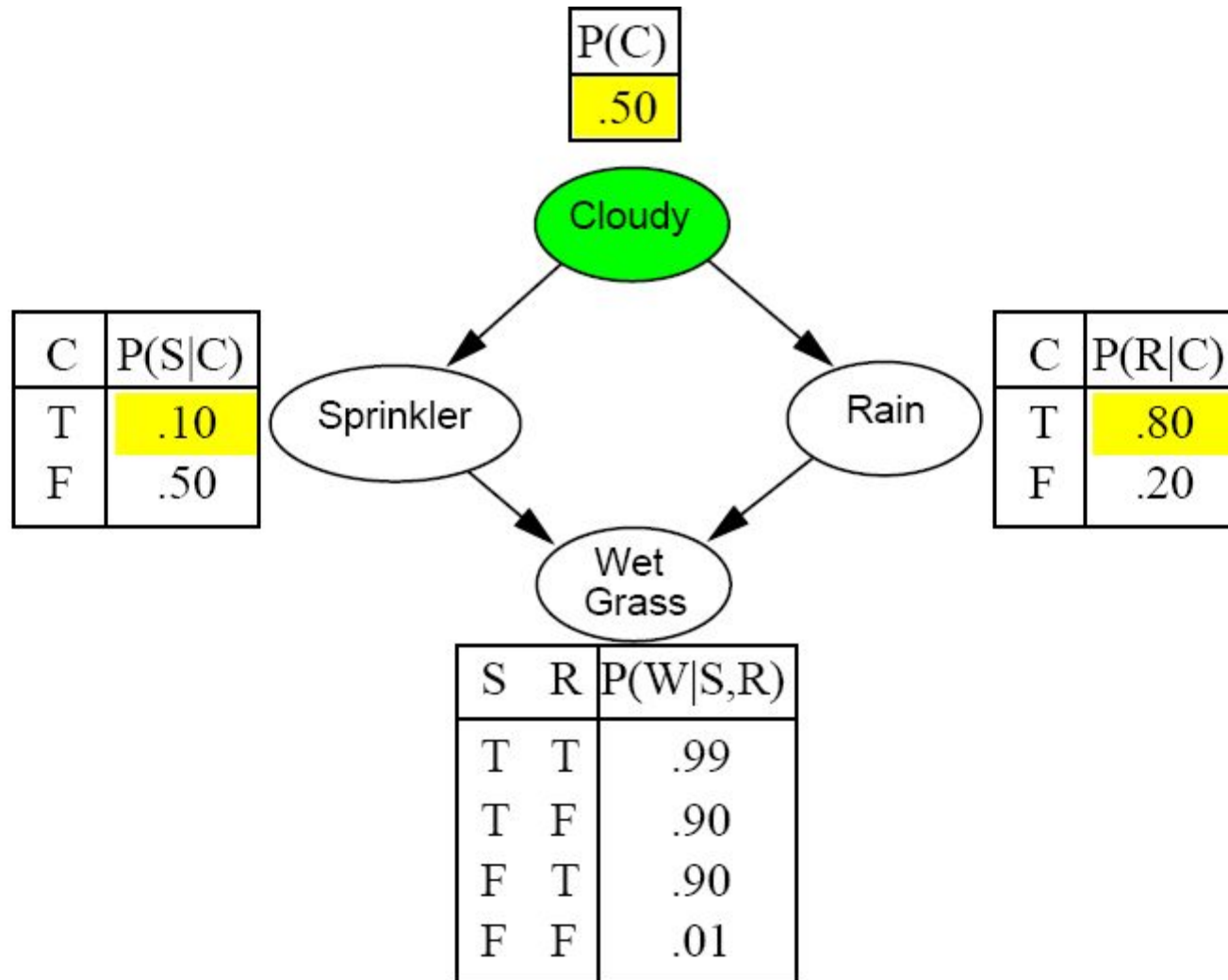
Variable order

Example: Sampling from a Bayesian Network

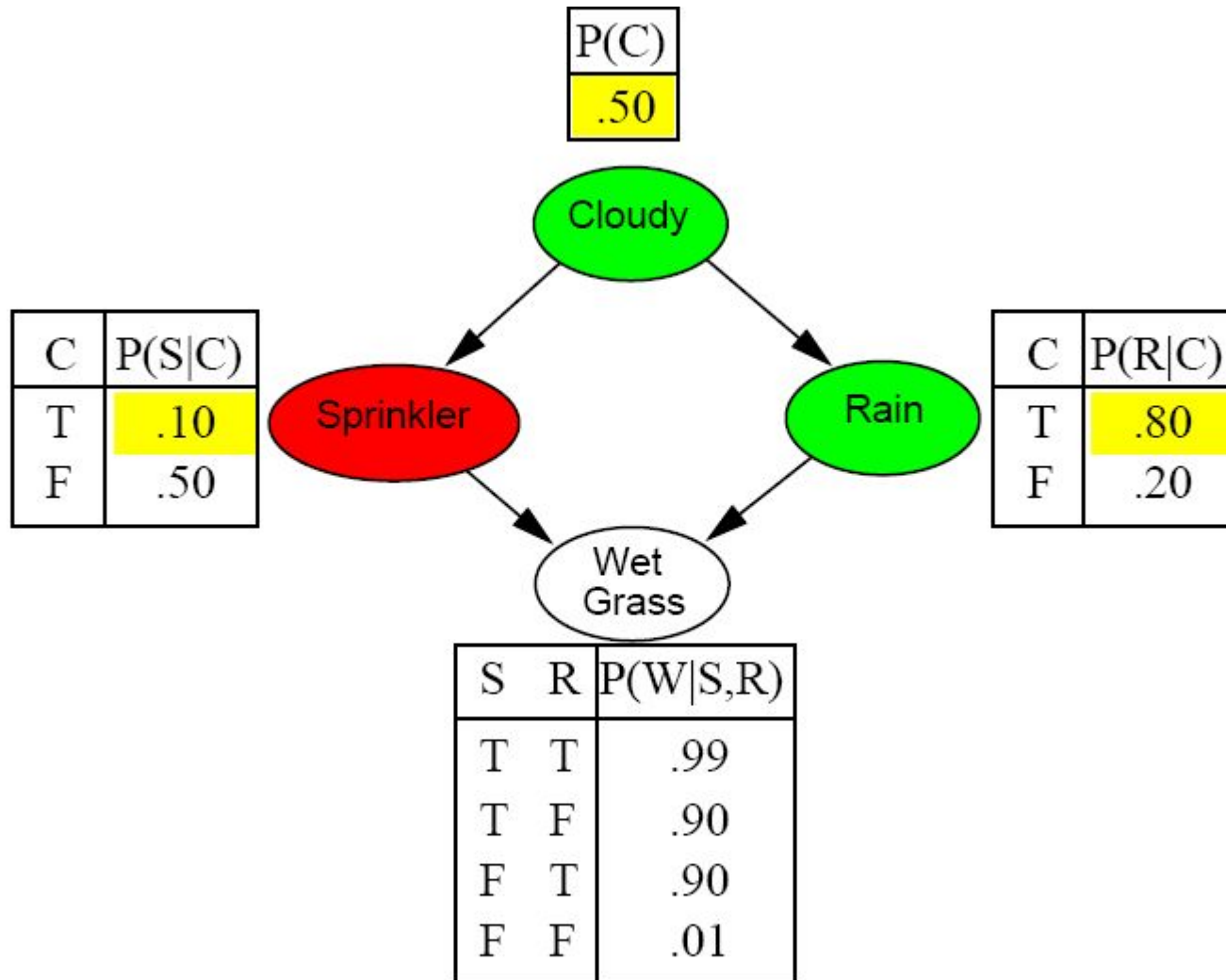
Draw a random value using the probability.



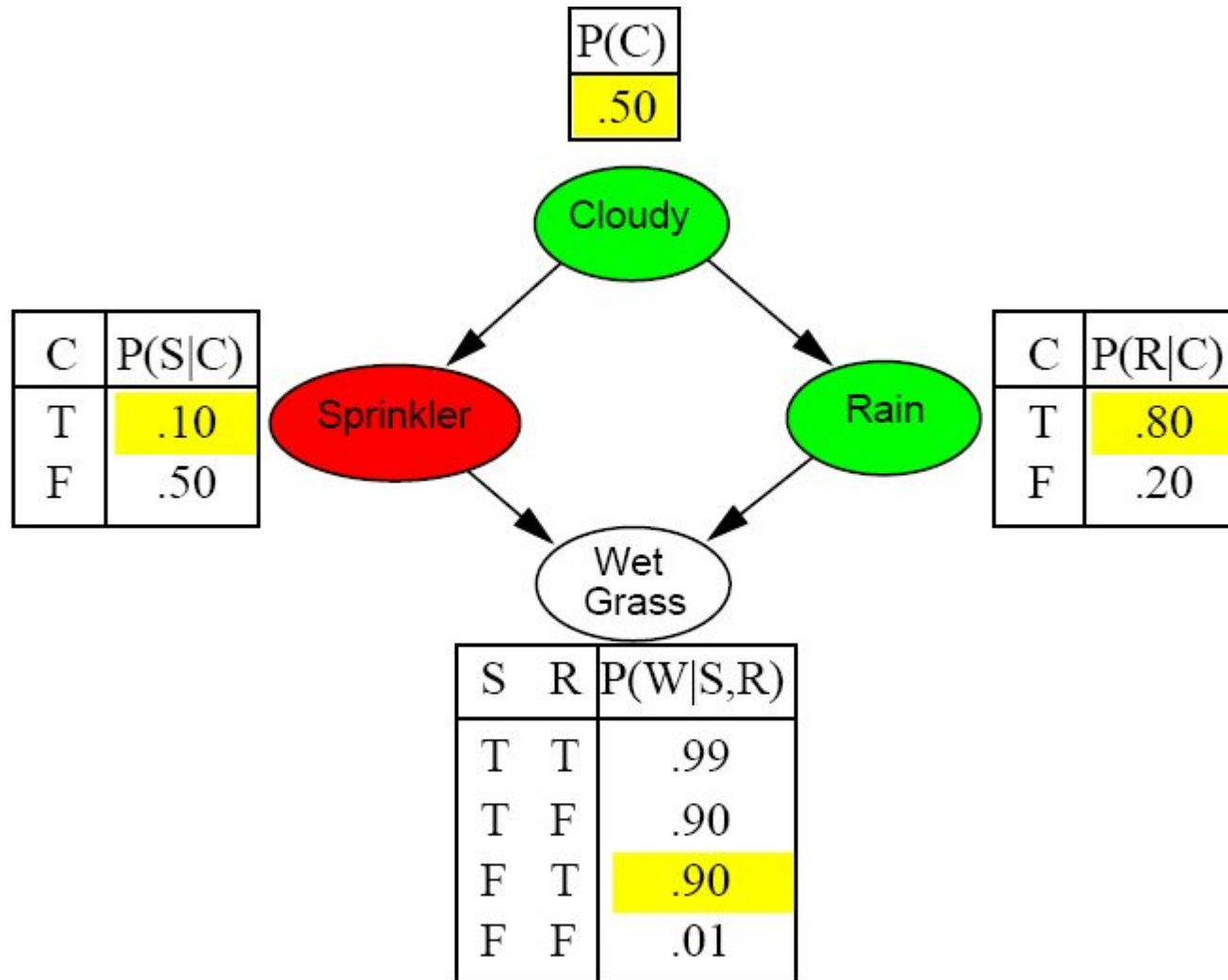
Example: Sampling from a Bayesian Network



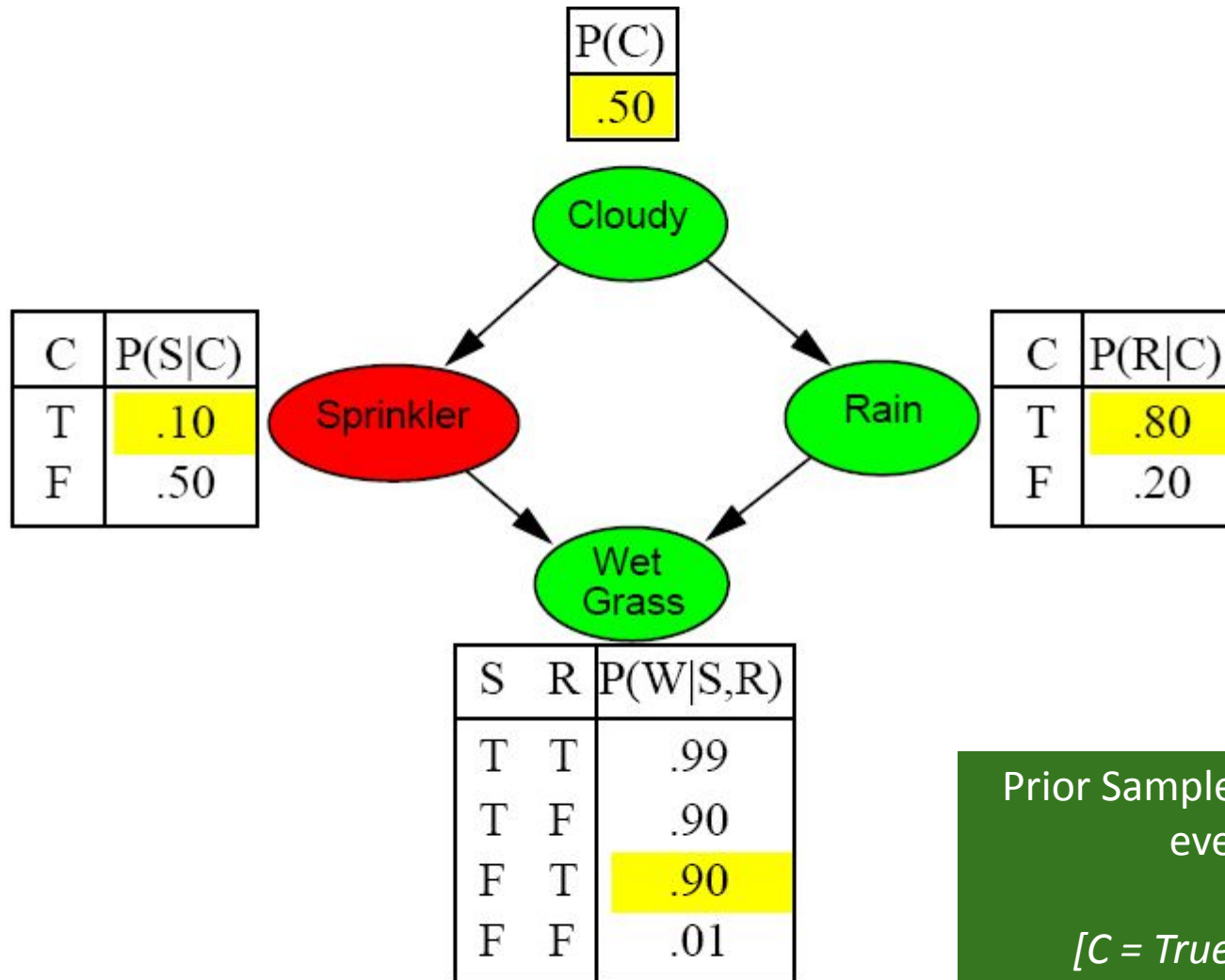
Example: Sampling from a Bayesian Network



Example: Sampling from a Bayesian Network



Example: Sampling from a Bayesian Network



Prior Sample returns the event:

$[C = \text{True}, S = \text{False}, R = \text{True}, W = \text{True}]$

Estimating the Joint Probability Distribution from Individual Samples

Sample N times and determine $N_{PS}(x_1, x_2, \dots, x_n)$, the count of how many times Prior-Sample produces event $(x_1, x_2, \dots, x_n)$.

$$\hat{P}(x_1, x_2, \dots, x_n) = \frac{N_{PS}(x_1, x_2, \dots, x_n)}{N}$$

The marginal probability of partially specified event (some x values are known) can also be calculated using the same samples. E.g.,

$$\hat{P}(x_1) = \frac{N_{PS}(x_1)}{N}$$

Estimating Conditional Probabilities: Rejection Sampling

Sample N times and **ignore the samples that are not consistent with the evidence e .**

$$\hat{P}(X|e) = \alpha N_{PS}(X, e) = \frac{N_{PS}(X, e)}{N_{PS}(e)}$$

Issue: What if e is a rare event?

- Example: burglary $\wedge$ earthquake
- Rejection sampling ends up throwing away most of the samples. This is very inefficient!

Estimating Conditional Probabilities: Rejection Sampling

```
function REJECTION-SAMPLING( $X, \mathbf{e}, bn, N$ ) returns an estimate of  $\mathbf{P}(X \mid \mathbf{e})$   
  inputs:  $X$ , the query variable  
            $\mathbf{e}$ , observed values for variables  $\mathbf{E}$   
            $bn$ , a Bayesian network  
            $N$ , the total number of samples to be generated  
  local variables:  $\mathbf{C}$ , a vector of counts for each value of  $X$ , initially zero  
  
  for  $j = 1$  to  $N$  do  
     $\mathbf{x} \leftarrow \text{PRIOR-SAMPLE}(bn)$   
    if  $\mathbf{x}$  is consistent with  $\mathbf{e}$  then  
       $\mathbf{C}[j] \leftarrow \mathbf{C}[j] + 1$  where  $x_j$  is the value of  $X$  in  $\mathbf{x}$   
  return NORMALIZE( $\mathbf{C}$ )
```

We throw away many samples
if $\mathbf{e}$ is rare!

Estimating Conditional Probabilities: Importance Sampling (likelihood weighting)

Goal: Avoid throw out samples like in rejection sampling.

1. Fix the evidence $E = e$ for sampling and estimate the probability for the non-evidence variables using prior-sampling. We call this probability $Q_{WS}(x)$

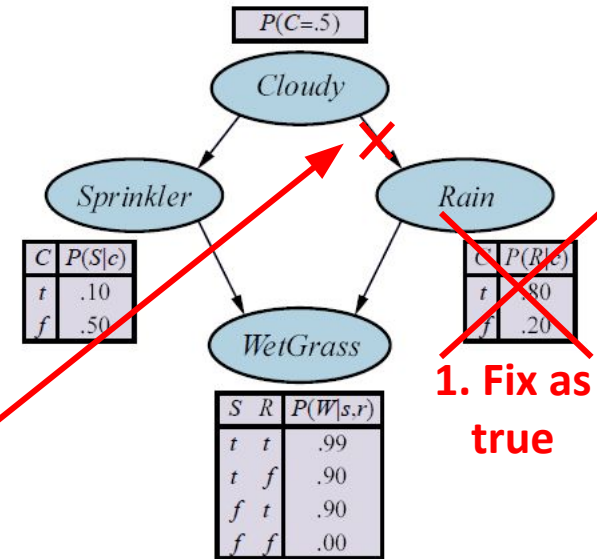
Note: Fixing the evidence breaks the dependence of the evidence variable on the evidence parents!

2. Correct the probabilities using weights $P(x|e) = w(x)Q_{WS}(x)$

The right weight is the chance that we see the evidence given its parent (to fix the broken dependence).

$$w(x) = \frac{1}{P(e)} \prod_{i=1}^m P(e_i | \text{parents}(E_i))$$

Example: Evidence = it rains



Estimating Conditional Probabilities: Markov Chain Monte Carlo Sampling (MCMC)

- **Idea:** Instead of creating each sample individually from scratch, **generates a sequence of samples**.
- Create the next state (= sample) by making random changes to the current state (= modify non-evidence variables). The sequence of states forms a random process called a **Markov Chain** (MC).
- The MC's stationary distribution turns out to be the posterior distribution of the non-evidence variables.
- The stationary distribution of a MC can be estimated using **Monte Carlo** simulation by counting how often each state is reached and normalize to obtain probability estimates.
- Algorithms:
 1. Gibbs sampling works well for BNs since we have CPTs.
 2. Metropolis-Hastings sampling is more general.

Note: Simulated annealing local search belongs to the family of MCMC algorithms.

Gibbs Sampling in Bayes Networks

```
function GIBBS-ASK( $X, \mathbf{e}, bn, N$ ) returns an estimate of  $\mathbf{P}(X | \mathbf{e})$   
  local variables:  $\mathbf{C}$ , a vector of counts for each value of  $X$ , initially zero  
     $\mathbf{Z}$ , the nonevidence variables in  $bn$   
     $\mathbf{x}$ , the current state of the network, initialized from  $\mathbf{e}$   
  
  initialize  $\mathbf{x}$  with random values for the variables in  $\mathbf{Z}$   
  for  $k = 1$  to  $N$  do  
    choose any variable  $Z_i$  from  $\mathbf{Z}$  according to any distribution  $\rho(i)$   
    set the value of  $Z_i$  in  $\mathbf{x}$  by sampling from  $\mathbf{P}(Z_i | mb(Z_i))$   
     $\mathbf{C}[j] \leftarrow \mathbf{C}[j] + 1$  where  $x_j$  is the value of  $X$  in  $\mathbf{x}$   
  return NORMALIZE( $\mathbf{C}$ )
```

Start with a random state

Change one non-evidence variable at a time

Count

- $mb(Z_i)$ is the Markov blanket of random variable Z_i . It makes sure that the new value is consistent with the other values. The Markov blanket of a variable consists of all variables it can be dependent of (parents, children and parents of children).

$$P(z_i | mb(Z_i)) = \alpha P(z_i | parents(Z_i)) \prod_{Y_i \in children(X_i)} P(y_i | parents(Y_j))$$

Gibbs Sampling: Example

Find

$$P(\text{Rain} \mid \text{Sprinkler} = \text{true}, \text{WetGrass} = \text{true}).$$

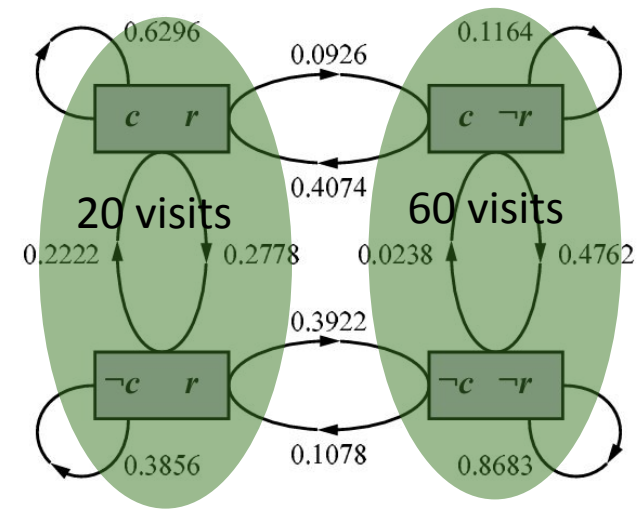
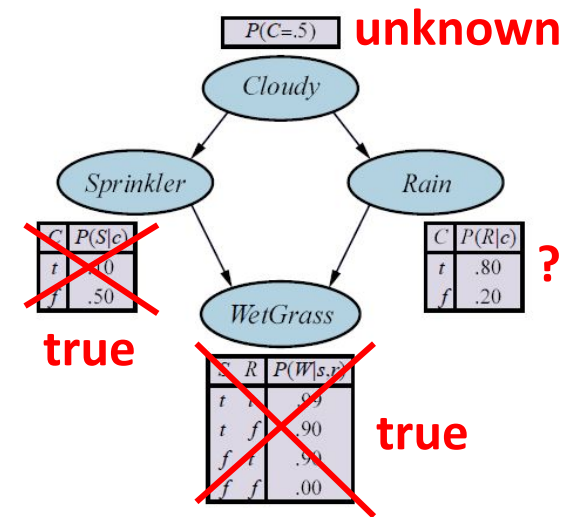
Determine states and calculate transition probabilities of the Markov chain for changing one variable using $P(z_i \mid mb(Z_i))$.

The algorithm randomly wanders around in this graph using the stated transition probabilities.

Assume that we observe 20 states with $\text{Rain} = \text{true}$ and 60 with $\text{rain} = \text{false}$:

$$\text{NORMALIZE}(\langle 20, 60 \rangle) = \langle 0.25, 0.75 \rangle$$

$$P(\text{Rain} \mid \text{Sprinkler} = \text{true}, \text{WetGrass} = \text{true}) \approx 0.25$$



Note the self-loops: the state stays the same when the resampled value is same it already has.



Conclusion

- Bayesian networks provide an efficient way to store a complete probabilistic model by exploiting (conditional) independence between variables.
- Inference means querying the model for a conditional probability given some evidence.
- Exact inference is difficult, for all but tiny models.
- State of the art is to use approximate inference by sampling from the model.
- Software libraries provide general inference engines.